

TH-RAG : Topic-Based Hierarchical Knowledge Graphs for Robust Multi-hop Reasoning in Graph-based RAG Systems

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Abstract

Retrieval-augmented generation (RAG) enables large language models (LLMs) to incorporate external knowledge at inference. Graph-based RAG extends this by organizing corpora into knowledge graphs, improving multi-hop reasoning and offering a global understanding of the corpus. However, triplet-based graphs generated by LLMs are often fragmented and poorly connected, which reduces coherence and hinders reasoning. Prior enrichment methods such as clustering, community detection, or approximate graph algorithms attempt to restore connectivity but incur high computational cost and risk semantic distortion. To address these issues, we propose TH-RAG, a hierarchical framework that organizes triplets into subtopics and topics, enhancing connectivity, integrating dispersed information, and supporting robust multi-hop reasoning. Experiments on abstractive and specific QA benchmarks show that TH-RAG outperforms strong baselines in accuracy and robustness while remaining efficient, providing a scalable foundation for graph-based RAG systems. To support further research, we release our code in our [GitHub repository](#).

1 Introduction

In recent years, large language models (LLMs) have demonstrated outstanding performance across various natural language processing tasks (Achiam et al., 2023; Yang et al., 2025a; Matarazzo and Torlone, 2025), owing to their extended context windows and strong document understanding capabilities (Team et al., 2023; Guo et al., 2025). However, integrating new knowledge into LLMs typically requires iterative fine-tuning, which incurs significant computational costs, consumes time, and introduces the risk of catastrophic forgetting (Kirkpatrick et al., 2017; Luo et al., 2023).

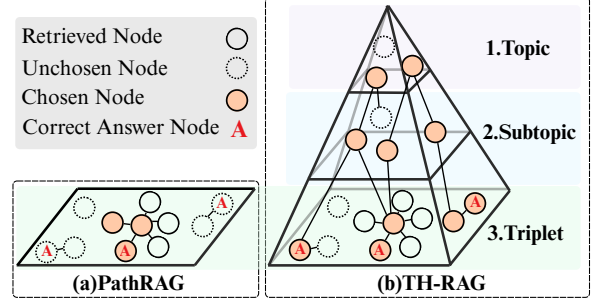


Figure 1: Simple example of TH-RAG compared to PathRAG. TH-RAG can efficiently retrieve almost all information in the corpus, since it utilizes a hierarchical graph structure.

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020; Gao et al., 2024) and graph-based RAG (Han et al., 2024) offer promising alternatives to overcome these challenges. RAG leverages sparse or dense retrieval mechanisms (Robertson and Zaragoza, 2009; Karpukhin et al., 2020) to fetch relevant information from external corpora and generates responses based on the retrieved content (Soudani et al., 2024; Balaguer et al., 2024).

Graph-based RAG further extends this paradigm by structuring the retrieval database as Knowledge Graphs (KGs) (Yang et al., 2025b; Kamra et al., 2024). This approach brings two key advantages: (1) it improves multi-hop reasoning over dispersed information compared to standard RAG, and (2) it provides a more global understanding of the corpus (Peng et al., 2024; Wu et al., 2024). Beyond these direct benefits, graph-based RAG also facilitates deeper document comprehension by capturing their logical structure and semantic relationships.

Recent graph-based RAG methods (Edge et al., 2024; Guo et al., 2024) have focused on constructing KGs directly from domain-specific corpora by extracting triplets (subject–relation–object). This fine-grained representation improves the precision of reasoning by structuring information at the semantic level.

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These methods often assume that sufficient connectivity exists among triplets within a chunk, which is rarely the case in practice (Han et al., 2024; Zhu et al., 2025c). However, in reality, triplet-based KGs generated by LLMs are frequently fragmented, leading to isolated subgraphs with poor inter-linkage. Such fragmentation not only reduces the overall semantic coherence of the graph but also prevents effective multi-hop reasoning, since critical relational paths between entities may be missing or disconnected. As a result, the model struggles to integrate dispersed information, ultimately hindering its ability to generate accurate and contextually grounded answers.

To address this limitation, prior studies have proposed graph enrichment techniques such as clustering-based summarization and community detection (Edge et al., 2024), or algorithms like HNSW (Xu et al., 2025; Wang et al., 2025). However, these methods often suffer from serious drawbacks: they introduce substantial computational overhead or distort detailed semantic relationships, which together compromise both the accuracy and reliability of the retrieved information.

To overcome these challenges, we propose **TH-RAG**, a novel graph-based RAG framework that constructs a **three-level hierarchical KG composed of Triplets, Subtopics, and Topics**. This semantic hierarchy enhances graph connectivity, facilitates integration across fragmented information, and supports efficient multi-hop reasoning and a global understanding of the corpus. TH-RAG operates in three stages: (1) **Hierarchical KG Construction**, where an LLM extracts Triplets, Subtopics, and Topics simultaneously to build a semantically structured graph; (2) **Topic-based Graph Traversal**, which begins from the most relevant Topic nodes and recursively explores related Subtopics and Entities to retrieve candidate Triplets; and (3) **Query-based Retrieval & Filtering**, where cosine similarity is computed between the query and each edge of candidate triplet, and the most relevant information is selected as the final context for answer generation.

Experimental results show that TH-RAG outperforms existing graph-based RAG methods across both abstractive (UltraDomain) and specific (MultiHop-RAG, HotpotQA) QA benchmarks. Additional ablation studies validate the effectiveness of our hierarchical graph design and retrieval strategy, demonstrating that TH-RAG offers a promising and scalable approach for enabling more robust

multi-hop reasoning in graph-based RAG systems.

2 Related Works

2.1 RAG and Graph-based RAG

Early RAG methods (Gao et al., 2024) utilized dense retrievers such as DPR (Karpukhin et al., 2020; Lewis et al., 2020) to chunk long documents into smaller units and retrieve relevant chunks based on similarity to a given query (Sharma, 2025; Hu and Lu, 2024; Gao et al., 2023). Since then, the RAG framework has evolved through integration with techniques such as reranking (Chen et al., 2024) or query expansion (Jagerman et al., 2023; Wang et al., 2023; Chan et al., 2024).

However, similarity-based retrieval alone often struggles with capturing logical dependencies or supporting multi-hop reasoning (Zhao et al., 2024; Wu et al., 2024). To address this limitation, graph-based RAG approaches have been proposed (Peng et al., 2024; Zhang et al., 2025b).

Initial graph-based RAG systems (Sun et al., 2023; Ma et al., 2024) relied on pre-constructed KGs such as Freebase (Bollacker et al., 2008) or Wikidata (Vrandečić and Krötzsch, 2014). More recent works have shifted toward constructing KGs directly from the corpus to improve adaptability to domain-specific settings (Zhu et al., 2024; Chen and Bertozzi, 2023).

2.2 Triplet-based Graph-based RAG

Triplet-based Graph-based RAG focuses on extracting triplets from within document chunks to build structured representations at the entity level (Han et al., 2024; Zhu et al., 2025c).

These triplet-based KGs are used to support structured multi-hop reasoning over the document content. Recent studies propose various enhancements to this paradigm, such as improving triplet connectivity (Luo et al., 2025), enabling lightweight reasoning (Luo et al., 2024; Böckling et al., 2025), or focusing on explicit path-based retrieval (Chen et al., 2025).

Edge et al. (2024) demonstrated promising results by constructing a triplet-based KG and applying community detection (Traag et al., 2019) to enhance semantic grouping and retrieval. Guo et al. (2024); Abane et al. (2024) proposed a more efficient and simplified usage of such graphs, relying on coarse graph structure for lightweight retrieval.

Subsequent studies (Liang et al., 2024; Jimenez Gutierrez et al., 2024; Gutiérrez et al.,

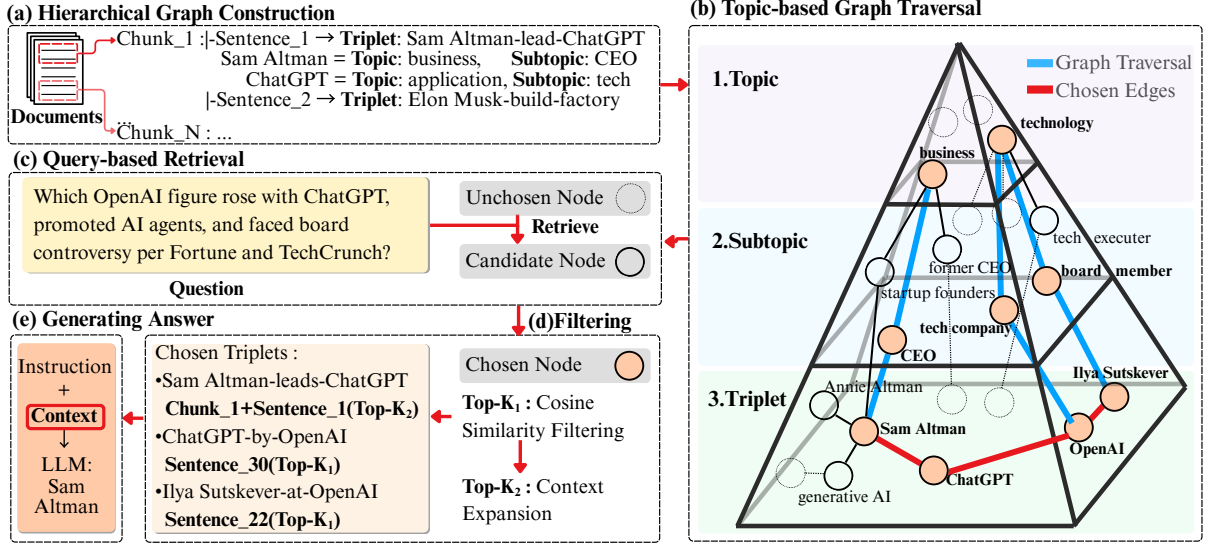


Figure 2: Overview of TH-RAG Framework. The framework consists of three stages: (a) **Hierarchical Graph Construction** – Documents are processed into triplet extraction with topics to build a hierarchical graph. (b) **Topic-based Graph Traversal** – The graph is navigated from topic to subtopic to triplet, guided by LLM-based relevance to the query. (c-d) **Query-based Retrieval & Filtering** – Triplets linked to selected subtopics are expanded by retrieving 1-hop neighboring entities. Retrieved triplets are filtered by cosine similarity (Top- K_1), and context is expanded from relevant chunks (Top- K_2).

2025) have explored various strategies to enhance graph connectivity and utility (Panda et al., 2024; Zhao et al., 2025). Some methods (Xu et al., 2025; Wang et al., 2025) adopt clustering techniques such as HNSW (Malkov, 2018) to group similar entities and reduce graph sparsity, while others employ explicit graph traversal strategies—such as path finding or reasoning over entity connections—to support multi-hop question answering (Luo et al., 2024; Han et al., 2025b; Chen et al., 2025; Böckling et al., 2025). In addition, several hybrid approaches (Zhu et al., 2025a; Sarmah et al., 2024) have been proposed that combine graph-based structures with traditional chunk-level retrieval.

3 Method

We now describe the architecture and implementation details of TH-RAG. Each component of the framework corresponds to the stages illustrated in Figure 2 with a simple example.

3.1 Hierarchical Graph Construction

Following prior graph-based RAG approaches, TH-RAG constructs a KG from a corpus by extracting triplets (subject–relation–object) using an LLM.

As discussed in Section 1, existing methods for addressing graph fragmentation face two key limitations: increased computational cost and impaired information fidelity.

To improve efficiency and minimize information distortion, we propose **Triplet Extraction with Topic**, a method that augments each extracted triplet with subtopic and topic annotations to form a hierarchically structured graph. Each entity is connected to one or more subtopics, and each subtopic to one or more topics, creating a semantic containment hierarchy:

- **Entities** represent factual units.
- **Subtopics** cluster semantically related entities.
- **Topics** abstract groups of subtopics into higher-level categories.

To ensure semantic grounding, we instruct the LLM to extract only corpus-relevant subtopics and topics (see Table 11). The output example can be found in Appendix E.1

Then we define the resulting graph as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where:

- $\mathcal{V}$ consists of three disjoint sets of nodes:
 - $\mathcal{V}_E$: entity nodes,
 - $\mathcal{ST}$: subtopic nodes,
 - $\mathcal{T}$: topic nodes.
- $\mathcal{E}$ consists of typed directed edges:
 - E_{triplet} : entity-to-entity relations (i.e., (s, r, o) where $s, o \in \mathcal{V}_E$),
 - E_{sub} : subtopic–entity links (i.e., (s, e) where $s \in \mathcal{ST}, e \in \mathcal{V}_E$),

- E_{top} : topic–subtopic links (i.e., (t, s) where $t \in T, s \in ST$).

Each edge in E_{triplet} also stores its source sentence as an attribute for sentence-level retrieval. As these sentences are directly extracted from the corpus, hallucination risk is minimized. These annotations support sentence-level retrieval in later stages. Moreover, graph updates require only one LLM call per chunk, making the method highly scalable.

Also, each entity is connected via at least two edges, and each subtopic is linked to both its entities and parent topic. This structure improves connectivity and prevents node isolation.

The overall process of graph construction is summarized in Algorithm 1.

Algorithm 1 Hierarchical Graph Construction

```

1: Input: Corpus chunk  $C$ 
2: Output: Hierarchical graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with triplets, subtopics, and topics
3: for each chunk  $c_i \in C$  do
4:    $\text{Triplets}, ST, T = \text{LLM}(C_i)$ 
5:   for each triplet  $(s, r, o) \in \text{Triplet}$  do
6:     Attach source sentence  $S$  to relation edge  $r$ 
7:     Connect  $s$  and  $o$  with edge  $r$ 
8:     Connect  $s$  and  $o$  to each  $st$ 
9:     Connect each  $st$  to its topic  $t$ 
10:  end for
11: end for

```

3.2 Topic-based Graph Traversal

To leverage the hierarchical structure of our graph, we design a two-step LLM-guided traversal strategy: **Topic-based Graph Traversal**.

Step 1: Topic Selection. Given a query, the LLM selects N_T relevant topics from all available topic nodes. Since topics are core keywords that represent the entire corpus, this step can be interpreted as the first step in defining the scope of the LLM’s response.

Step 2: Subtopic Selection. For each selected topic, the LLM chooses N_{ST} subtopics from its connected subtopic nodes, based on semantic relevance to the query. In practice, N_T and N_{ST} are bounded to small values, enabling our traversal method to scale with minimal LLM calls.

While the entire list of candidates is provided in each step, the extended context capacity of modern LLMs (Hurst et al., 2024) ensures that this selection process remains efficient. Typically, N_T and N_{ST}

are small values, requiring just one LLM call for topic selection and N_T calls for subtopic selection.

This approach offers greater robustness compared to methods that extract entities from the query (Guo et al., 2024) or implicitly infer topics and subtopics. By providing the LLM with explicit lists of candidate topics, subtopics, and entities as context, it selects the most relevant ones based on the query, reducing ambiguity and increasing reliability. Although topics and subtopics are generated by LLMs and may exhibit imperfect consistency, we mitigate this issue by allowing the selection of multiple candidates rather than enforcing a single choice, thereby ensuring broader coverage and stable retrieval.

Ultimately, this process can be viewed as a hierarchical graph traversal that progressively narrows down the search space within a large corpus to efficiently locate the answer (a visualization of results can be found in Figure 5.)

Algorithm 2 Topic-based Graph Traversal & Query-based Retrieval

```

1: Input: Query  $q$ , graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , parameters  $N_T, N_{ST}, K_1, K_2$ 
2: Output: Final context set
3: Extract top- $N_T$  topics relevant to  $q$ :
4:    $T_{\text{selected}} = \text{LLM}(T_{\text{list}}, q)$ 
5: for each  $t_i \in T_{\text{selected}}$  do
6:   Extract top- $N_{ST}$  subtopics relevant to  $t_i$  and  $q$ :
7:      $ST_i = \text{LLM}(ST_{\text{list}}^{(t_i)}, q)$ 
8:      $ST_{\text{selected}} \leftarrow ST_{\text{selected}} \cup ST_i$ 
9: end for
10: Retrieve all entities under  $ST_{\text{selected}}$ 
11: For each entity  $e$ , collect its 1-hop neighbors
12: Compute similarity between  $q$  and all sentences
13: Select top- $K_1$  sentences to form the primary context  $\mathcal{C}_{\text{primary}}$ 
14: From  $\mathcal{C}_{\text{primary}}$ , select the top- $K_2$  sentences and include their source chunks as extended context

```

3.3 Query-based Retrieval

From each selected subtopic, we collect the connected entity nodes, which act as anchors for context retrieval. For each entity, we explore its 1-hop neighbors within the graph, collecting all associated edges, since each edge is annotated with its original source sentence. This process yields a set of candidate evidence sentences directly grounded

in the source corpus.

These edge-level sentences form the basis for our filtering mechanism, enabling precise and faithful sentence-level evidence retrieval. To reduce redundancy and improve relevance, we apply a **two-stage filtering strategy**:

- **Cosine Similarity Filtering:** We compute the cosine similarity between the query and each candidate sentence. The top- K_1 most relevant sentences are selected as the primary context for generation.
- **Context Expansion:** We further select a subset of K_2 sentences ($K_2 \ll K_1$) and retrieve their full source chunks. This expansion provides additional contextual cues around high-confidence sentences.

As noted by Han et al. (2025a), answer entities or key supporting sentences are sometimes omitted during the graph construction process. To mitigate this risk, we adopt targeted context expansion around the most relevant sentences. Thus, TH-RAG uses chunk- and triplet-level information to generate answers. The overall process of topic-based graph traversal and query-based retrieval is summarized in Algorithm 2.

3.4 Multi-hop Reasoning Robustness

Graph-based RAG approaches face two major challenges:

Graph fragmentation: When triplets are sparsely connected, reasoning paths between related entities are easily broken. For example, from the sentence “Marie Curie and Pierre Curie conducted groundbreaking research on radioactivity together”, an extractor may only produce (Marie Curie, conducted, research), omitting Pierre Curie. This omission (Edge et al., 2024; Han et al., 2025a) eliminates the collaboration link and prevents correct multi-hop reasoning.

High computational cost and risks of graph enrichment: Many methods (Edge et al., 2024; Wang et al., 2025; Xu et al., 2025) construct hierarchical structures through summarization (e.g., clustering or community detection) or graph algorithms such as HNSW (Malkov, 2018). However, these approaches are computationally expensive and time-consuming, and summarization-based techniques risk losing important information or even introducing hallucinated content during the process.

To mitigate these issues, TH-RAG constructs a three-level hierarchical graph (Topics, Subtopics,

Dataset	Tokens	Passages	# QA	Tokens/P
Agri	1.9M	12	125	158K
CS	2.0M	10	125	200K
Legal	4.7M	94	125	50K
Mix	602K	61	125	9.9K
Hotpot	1.2M	9,827	1,000	122
MultiHop	991K	435	1,000	2.3K

Table 1: **Document statistics** for our experimental datasets. Agri, Hotpot, and MultiHop refer to Agriculture, HotpotQA, and MultiHop-RAG, respectively. Tokens/P indicates the average number of tokens per passage.

and Entities) with only a single LLM call per chunk, thereby achieving near-zero additional enrichment cost while improving connectivity. Furthermore, targeted context expansion retrieves additional evidence sentences around high-confidence nodes, reducing the risk of entity omission. Most importantly, every edge in TH-RAG retains its original source sentence as an attribute, ensuring that no semantic information from the corpus is lost and that retrieval can always fall back to faithful sentence-level evidence rather than relying solely on the quality of abstracted triplets.

Taken together, these strategies enable TH-RAG to maintain robustness in multi-hop reasoning: narrowing the search space to the region likely to contain the correct answer, utilizing surrounding information while preventing loss of corpus data. We further analyze the consistency and semantic validity of the generated topics and subtopics in Appendix D.2.

4 Experiments

To evaluate the effectiveness and robustness of TH-RAG, we design experiments to answer the following research questions(RQs):

- **RQ1:** Is our method effective for QA datasets of multi-hop reasoning and global understanding?
- **RQ2:** How well does our method perform, especially in terms of mitigating graph fragmentation?
- **RQ3:** How efficient is our method in terms of resource usage and scalability?
- **RQ4:** What are the core components of our method and the optimal hyperparameters?

4.1 Datasets

In our experiments, we used two types of data sets. One is an **abstractive QA** dataset, such as **UltraDomain** (Qian et al., 2025), which requires

	Agriculture								CS							
	Comprehensive		Diversity		Empowerment		Overall		Comprehensive		Diversity		Empowerment		Overall	
	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline
Win Rate	84.2%	15.8%	88.3%	11.7%	87.5%	12.5%	86.7%	13.3%	86.9%	13.1%	91.0%	9.0%	86.9%	13.1%	86.9%	13.1%
vs Naive	87.0%	13.0%	91.1%	8.9%	88.6%	11.4%	88.6%	11.4%	78.7%	21.3%	74.6%	25.4%	78.7%	21.3%	78.7%	21.3%
vs GraphRAG G	88.7%	11.3%	90.3%	9.7%	89.5%	10.5%	89.5%	10.5%	84.6%	15.4%	86.2%	13.8%	87.0%	13.0%	86.2%	13.8%
vs GraphRAG L	87.1%	12.9%	91.9%	8.1%	89.5%	10.5%	88.7%	11.3%	80.7%	19.3%	80.7%	19.3%	81.5%	18.5%	81.5%	18.5%
vs LightRAG	80.7%	19.3%	92.4%	7.6%	85.7%	14.3%	85.7%	14.3%	78.4%	21.6%	86.4%	13.6%	81.6%	18.4%	81.6%	18.4%
vs PathRAG	52.3%	47.7%	60.4%	39.6%	51.4%	48.6%	52.3%	47.7%	49.1%	50.9%	46.4%	53.6%	50.9%	49.1%	49.1%	50.9%
vs HypergraphRAG																
	Legal								Mix							
	Comprehensive		Diversity		Empowerment		Overall		Comprehensive		Diversity		Empowerment		Overall	
	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline	TH-RAG	Baseline
Win Rate	86.2%	13.8%	89.4%	10.6%	90.2%	9.8%	90.2%	9.8%	91.9%	8.1%	95.5%	4.5%	93.7%	6.3%	93.7%	6.3%
vs Naive	79.8%	20.2%	69.4%	30.6%	81.5%	18.5%	81.5%	18.5%	84.5%	15.5%	84.1%	15.9%	90.1%	9.9%	90.1%	9.9%
vs GraphRAG G	89.4%	10.6%	87.0%	13.0%	90.2%	9.8%	90.2%	9.8%	96.5%	3.5%	98.3%	1.7%	96.5%	3.5%	96.5%	3.5%
vs GraphRAG L	85.5%	14.5%	85.5%	14.5%	89.5%	10.5%	89.5%	10.5%	91.3%	8.7%	95.7%	4.3%	92.2%	7.8%	92.2%	7.8%
vs LightRAG	86.4%	13.6%	84.0%	16.0%	86.4%	13.6%	86.4%	13.6%	90.5%	9.5%	97.4%	2.6%	92.2%	7.8%	92.2%	7.8%
vs PathRAG	50.9%	49.1%	43.8%	56.2%	50.9%	49.1%	50.9%	49.1%	57.9%	42.1%	63.2%	36.8%	57.0%	43.0%	57.9%	42.1%
vs HypergraphRAG																

Table 2: **Main Results on UltraDomain**, specifically for Agriculture, CS, Legal and Mix domains. Metrics are reported using 1-vs-1 win rates under an LLM-as-a-judge setting. We exclude GraphRAG-G from our retrieval evaluation, as its use of global community detection and summarization spans numerous chunks, making the comparison less meaningful in our setting.

	Answer Quality								Retrieval Quality							
	MultiHop-RAG				HotpotQA				MultiHop-RAG				HotpotQA			
	F1	Precision	Recall	Accuracy	F1	Precision	Recall	Accuracy	Recall	F1	Rec@5	NDCG@5	Recall	F1	Rec@5	NDCG@5
Naive	0.501	0.475	0.599	0.604	0.584	0.612	0.590	0.509	0.330	0.210	0.337	0.375	0.705	0.175	0.693	0.662
GraphRAG-G	<u>0.526</u>	0.501	0.618	<u>0.653</u>	0.393	0.410	0.402	0.343	-	-	-	-	-	-	-	-
GraphRAG-L	0.469	0.451	0.536	0.535	0.668	<u>0.696</u>	<u>0.678</u>	0.595	0.267	0.239	0.267	0.412	0.830	0.479	0.833	0.794
LightRAG	0.464	0.448	0.527	0.526	0.496	0.519	0.507	0.439	0.072	0.039	0.061	0.082	0.323	0.129	0.282	0.217
PathRAG	0.468	0.453	0.523	0.525	0.551	0.578	0.562	0.488	0.203	0.113	0.182	0.265	0.818	0.326	0.808	<u>0.805</u>
HyperGraphRAG	<u>0.526</u>	<u>0.503</u>	<u>0.619</u>	0.621	0.674	0.703	0.683	<u>0.599</u>	0.426	0.283	<u>0.402</u>	<u>0.460</u>	<u>0.848</u>	0.382	<u>0.848</u>	0.763
TH-RAG	0.712	0.711	0.720	0.722	<u>0.671</u>	<u>0.692</u>	0.685	0.612	<u>0.405</u>	<u>0.271</u>	0.405	0.555	0.886	<u>0.383</u>	0.886	0.893

Table 3: **Main results on HotpotQA and MultiHop-RAG**. **Bold** indicates the best result, and underline indicates the second-best. Rec@5 refers to the Recall at 5 metric.

answering open-ended questions based on broad knowledge. Following prior studies (Edge et al., 2024; Guo et al., 2024), we used three domain-specific datasets (Agriculture, CS, and Legal) and one mixed-domain corpus, from which we generated 125 questions.

The other is a **specific QA** dataset, **HotpotQA** (Yang et al., 2018) and **MultiHop-RAG** (Tang and Yang, 2024), which has concrete multi-hop evidence that must be retrieved to generate answers. More detailed explanations about datasets are provided in Appendix C.1

We randomly selected 1,000 QA pairs along with their corresponding passages from both MultiHop-RAG and HotpotQA to construct the corpus for evaluation. A detailed description of these datasets is provided in Table 1.

4.2 Metrics

We used two evaluation approaches: For the **UltraDomain** dataset, we applied the **LLM-as-a-judge** method (Zheng et al., 2023), comparing answers pairwise as in Guo et al. (2024). For **MultiHop-RAG** and **HotpotQA**, we used traditional metrics—**F1**, **Recall**, **Precision**, and **Accuracy**—along with retrieval metrics like **recall**, **F1**, **recall@5**, and **NDCG@5**. A detailed description of these metrics is provided in Appendix C.4.

4.3 Baselines

We compared TH-RAG against several representative baselines categorized into three groups: (1) a basic retrieval form, **NaiveRAG** (Gao et al., 2024); (2) triplet-based graph baselines, including **GraphRAG** (Edge et al., 2024) and **LightRAG** (Guo et al., 2024); and (3) current state-of-the-art methods, **PathRAG** (Chen et al., 2025) and **HyperGraphRAG** (Luo et al., 2025). For GraphRAG, we implemented both the local and global retrieval methods. We refer to the local version as **GraphRAG-L** and the global version as **GraphRAG-G** throughout our experiments. A detailed explanation of baselines can be found in Appendix C.3 and additional experiment on HippoRAG2 (Gutiérrez et al., 2025) can be found in Appendix D.3.

4.4 Implementation Details

We used the following hyperparameters and implementation settings: K_1 and K_2 were fixed at 30 and 5, respectively. N_T and N_{ST} were determined through prompt-based selection, with values ranging from 5–10 and 10–25, respectively. The exact numbers varied depending on the LLM’s output. Additional implementation details are in Appendix B, and used prompts are in Appendix A.

	LightRAG	TH-RAG
# Nodes	20,914	50,162
# Topic Nodes	-	424
# Subtopic Nodes	-	14,319
# Entity Nodes	20,914	35,419
# Edges	24,707	94,507
# Topic-subtopic edges	-	19,212
# Subtopic-entity edges	-	41,268
# Entity-Entity edges	34,027	21,906
# Subgraphs	8,805	3
% Biggest Subgraph	56.11%	99.98%

Table 4: Constructed graph statistics comparison with LightRAG on Legal dataset. The difference in the number of entities between the two methodologies is due to LightRAG’s entity merging and deletion. The percentage of subgraph is calculated on nodes.

5 Results

5.1 Main Results (RQ1)

On the **UltraDomain** dataset, TH-RAG surpasses all baselines except HyperGraphRAG across all four domains (Table 2).

- Against **PathRAG**, a widely recognized state-of-the-art method, TH-RAG achieves an **average win rate of 86.48%**, highlighting its strong competitiveness in global understanding of the corpus.
- While HyperGraphRAG slightly outperforms in the CS domain, TH-RAG consistently dominates in Agriculture, Legal, and Mix, demonstrating stable cross-domain generalization.
- In the **mixed-domain setting**, TH-RAG exhibits the most pronounced gap, underlining its robustness in handling diverse, open-domain QA.

On the **specific-type** datasets, TH-RAG consistently achieves **state-of-the-art scores** across retrieval and reasoning metrics (Table 3).

- In **MultiHop-RAG**, it surpasses GraphRAG-G and HyperGraphRAG by **6.9%** and **10.1%**, respectively, demonstrating clear superiority in multi-hop reasoning.
- These gains confirm that TH-RAG’s hierarchical, top-down retrieval strategy is highly effective in multi-hop and fact-intensive QA.

Importantly, TH-RAG is not restricted by dataset characteristics. It performs consistently well regardless of passage length, and it shows robustness across both **abstractive-type** and **specific-type** QA tasks. This generalization ability highlights that the model’s design is inherently adaptive, enabling strong performance. An additional observation is that NaiveRAG performs unexpectedly well on specific-type datasets. This suggests that errors in graph construction—such as missing entities—can

	TH-RAG	HyperGraphRAG
Comparison	0.625	0.538
Temporal	0.509	0.197
Inference	0.954	0.938
Null	0.786	0.664

Table 5: Comparison with HyperGraphRAG by question type on the MultiHop-RAG dataset.

critically harm performance (Edge et al., 2024; Han et al., 2025a).

5.2 Graph Fragmentation and Robustness Analysis (RQ2)

To assess the impact of TH-RAG’s hierarchical structure on mitigating graph fragmentation, we compare the structural properties of graphs constructed by TH-RAG and a representative triplet-based method, LightRAG.

Compared to LightRAG, TH-RAG substantially reduces the number of disconnected subgraphs and achieves a much higher largest-connected-component ratio. These improvements highlight the effectiveness of our Topic–Subtopic–Entity hierarchy in enhancing global graph connectivity. Summary statistics are presented in Table 4. Results on other datasets are provided in Appendix C.2, and visualization results are included in Appendix E.3.

Figure 1 further illustrates the benefit of reduced fragmentation through a qualitative comparison with PathRAG. In conventional triplet-based methods, answer-relevant entities often appear in separate subgraphs, making reasoning paths incomplete or unreachable—especially for methods like PathRAG that depend heavily on connectivity. In contrast, TH-RAG does not rely on direct entity–entity connections. Instead, it accesses relevant information by navigating through topic-based hierarchical graph traversal and retrieving sentence-level evidence, enabling robust reasoning even in partially disconnected entities.

Furthermore, the number of topic nodes remains small, and the average Topic-to-Subtopic ratio is approximately *1:30*. This ensures that the Topic and Subtopic Selection process remains token-efficient and computationally lightweight during inference.

We also provide a comparison of question-type-level performance on MultiHop-RAG in Table 5. While TH-RAG performs comparably to HyperGraphRAG on *Inference*, *Comparison*, and *Null* types, it significantly outperforms on *Temporal* questions. This suggests that TH-RAG’s sentence-based retrieval and topic-aware traversal are better

	Light	Hyper	Local	TH-RAG
Indexing Call	5,978	2,772	4,354	902
Indexing Token	8M	20.3M	15M	2.3M
Querying Time	2.66s	9.78s	0.77s	3.54s
Context Token	25K	20K	13.6K	7.4K

Table 6: Efficiency comparison of representative baseline methods on MultiHop-RAG. Token counts include both prompt and context. Light, Hyper, and Local refer to LightRAG, HyperGraphRAG, and GraphRAG-L, respectively.

	Accuracy	F1	Recall	Precision
Original	0.722	0.712	0.72	0.71
w/o chunks	0.580	0.576	0.577	0.577
w/o Triplets	0.692	0.68	0.691	0.678
w/o Traversal	0.624	0.62	0.622	0.62

Table 7: Ablation study on key components of TH-RAG. W/o Traversal means we don’t apply graph-traversal, using only filtering by all sentences.

at capturing temporally grounded relations compared to HyperGraphRAG, leading to high robustness of TH-RAG.

5.3 Efficiency Analysis (RQ3)

We next evaluate the efficiency of TH-RAG in terms of token usage and LLM call overhead, focusing on two key stages: indexing and retrieval. We compare TH-RAG against **GraphRAG-L**, **HyperGraphRAG**, and **LightRAG**—three strong baselines known for either high performance or retrieval efficiency (Table 6).

In the indexing phase, TH-RAG demonstrates remarkable efficiency. It requires only **32.5%** of the LLM calls used by HyperGraphRAG (902 vs. 2,772) and just **29%** of the tokens consumed by LightRAG (2.3M vs. 8M). This reduction is primarily attributed to our graph construction method and prompt-based topic/subtopic annotation, which eliminate the need for costly iterative clustering or summarization at the entity level.

During retrieval, TH-RAG incurs slightly higher latency compared to GraphRAG-L due to the $(N_T + 1)$ LLM calls needed for topic and subtopic selection. Nevertheless, its total token usage remains low—only **54%** of that required by HyperGraphRAG (7.4K vs. 13.6K). Since the number of topic nodes rarely exceeds 1,000, the retrieval time complexity remains $O(N_T)$, making TH-RAG scalable even for large corpora. Overall, TH-RAG achieves a favorable balance between computational efficiency and retrieval quality.

	$K_1=5$	10	30	50
$K_2=1$	0.536	0.565	0.622	0.630
3	0.662	0.679	0.697	0.702
5	0.697	0.685	0.72	0.719
10	-	0.714	0.743	0.726

Table 8: Ablation on K_1 & K_2 on MultiHop-RAG with accuracy. Since K_2 cannot be greater than K_1 by definition, that case is marked as -

5.4 Ablation and Hyperparameter Analysis (RQ4)

We conduct ablation studies to evaluate the contribution of each component in TH-RAG. As shown in Table 7, removing chunks in context leads to significant performance degradation. Removing triplet-level information or bypassing the topic-subtopic traversal (e.g., applying filtering over all sentences) also results in noticeable accuracy drops.

These results confirm that TH-RAG’s strength lies in its ability to semantically scope the graph through topic and subtopic selection, enabling it to isolate focused subgraphs that are rich in relevant information. This targeted traversal leads to the extraction of high-quality chunks grounded in the original corpus, enabling more robust and reliable multi-hop reasoning.

We also evaluate the impact of varying the hyperparameters K_1 and K_2 , which control the number of retrieved sentences and the number of expanded chunks, respectively (Table 8). While our main experiments adopt $K_1 = 30$ and $K_2 = 5$ for cost efficiency, increasing K_2 to 10 leads to slightly better performance, indicating a trade-off between answer quality and token cost (Joren et al., 2024a).

Interestingly, increasing both K_1 and K_2 beyond a certain point (e.g., $K_1 = 50$ and $K_2 = 10$) degrades performance—likely due to *context rot* or *lost-in-the-middle* effects, as noted in recent studies (Zhang et al., 2025a; Hsieh et al., 2024). This highlights the importance of careful context engineering (Mei et al., 2025; Joren et al., 2024b) and hyperparameter tuning in retrieval-augmented systems.

6 Conclusion

We proposed TH-RAG, a novel graph-based RAG framework designed to address two central challenges of prior methods: (1) reduced utility of graph structures due to graph fragmentation, and (2) information loss and excessive compu-

tational cost introduced by graph enrichment techniques. To mitigate these issues, TH-RAG constructs a three-level hierarchical knowledge graph—composed of topics, subtopics, and entities—that semantically organizes information extracted from unstructured text. Our method improves connectivity with near-zero additional cost, while preserving the original corpus information. A topic-guided retrieval strategy further enables the model to operate directly on sentence-level evidence, ensuring robust multi-hop reasoning.

Through this integrated approach, TH-RAG consistently demonstrates robust performance across diverse datasets—ranging from long-passage to short-passage and abstractive QA—highlighting its general applicability. This robustness distinguishes TH-RAG from other methods and underscores its effectiveness as a reliable and extensible foundation for graph-based RAG.

Limitations

TH-RAG introduces a hierarchical KG built from LLM-extracted topics, subtopics, and triplets. However, the current approach has several limitations that suggest avenues for future improvement. First, the topic and subtopic normalization remains imperfect. Due to inconsistencies in LLM outputs, semantically similar concepts are often assigned to different topic or subtopic labels, unnecessarily inflating the graph structure (e.g., sports/sport, film director/director). We also tried setting up a predefined schema for topics and subtopics to ensure consistency, but we were unable to achieve satisfactory results. To address this, future work could explore embedding-based clustering techniques to group semantically equivalent nodes (Chang et al., 2025; Liu et al., 2025b). For more details on topic consistency in TH-RAG, see Appendix D.2. Second, this work deliberately omits widely-used RAG techniques such as query expansion and context reranking, in order to isolate the effectiveness of our hierarchical graph structure in its most basic and efficient form. However, given the demonstrated effectiveness of these techniques in recent literature (Gao et al., 2024; Sharma, 2025), integrating them in a way that aligns with our topic-based hierarchy could further enhance performance. Lastly, this method was evaluated exclusively using GPT-4o-mini, which is currently the most readily available high-performance LLM. Since TH-RAG directly utilizes the LLM for topic/subtopic selec-

tion, the LLM’s performance can directly impact TH-RAG’s performance. We conducted a simple test and found that, compared to GPT-4o-mini, selecting a better LLM improves TH-RAG’s performance, while selecting a weaker LLM degrades it. However, since the magnitude of this variation was similar to that of NaiveRAG, we omitted this section from the paper.

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Appendices

A Prompts

A.1 Answer Generation Prompt

Instruction description
—Role— You are a helpful assistant responding to user query
—Goal— Generate a concise response based on the following information and follow Response Rules. Do not include information not provided by following Information
—Target response length and format— Multiple Paragraphs
—Information— {{context}}
—Response Rules— - Use markdown formatting with appropriate section headings - Please respond in the same language as the user’s question. - If you don’t know the answer, just say so. - Do not make anything up. Do not include information not provided by the Information.
—Query— {{question}}

Table 9: Answer Generation Prompt for UltraDomain. This prompt is used when we need long, comprehensive response.

A.2 Short Answer Generation Prompt

Instruction description
—Role— You are a multi-hop retrieval-augmented assistant.
—Goal— Read the Information passages and generate the correct answer to the Query. Use only the given Information; if it is insufficient, reply with "Insufficient information.". If you need to answer like yes or no, use "Yes" or "No" only.
—Target response length and format— - One-word or minimal-phrase answer (max 5 words).
—Response Rules— - Answer must be short and concise. - Answer language must match the Query language. - Do NOT add or invent facts beyond the Information.
—Information— {{context}}
—Query— {{question}}

Table 10: Short Answer Generation Prompt used for HotpotQA and MultiHop-RAG. This prompt is used when we need short, concise response.

A.3 Triplet Extraction with Topic Prompt

Instruction description
—Role— You are a highly skilled information extraction system designed to process factual information accurately and clearly. —Goal— Extract factual (subject, relation, object) triples from the document and classify the subject and object into a subtopic and a main topic. —Instructions— 1. Read the entire document below and extract all factual (subject, relation, object) triples. Each triple must be grounded in a specific sentence from the document. 2. Paraphrasing is acceptable only if the relation is clearly implied by the sentence. 3. Resolve all pronouns such as "it", "he", "she", "they", etc. using the surrounding context. Replace all pronouns in the triple with their correct referents. - Do not include any unresolved or ambiguous pronouns in the triples. - Be specific and use full entity names instead of pronouns wherever applicable. 4. For each subject and object: - Assign a Subtopic (a specific category such as "Electronic Musician", "Sound Label", etc.) - Assign a Main topic (a broader category such as "Music", "Art", etc.) - Ensure the subtopic and main topic reflect both the entity and the overall context of the document. 5. Return only valid JSON in the specified format. Do not include markdown, comments, or any other text. 6. Ensure that the JSON is well-formed and valid. —Examples— {{example}} —Input Document— {{document}}

Table 11: Triplet Extraction with Topic Prompt

A.4 Topic Selection Prompt

Instruction description
—Goal— Given the user’s question, choose all topics from the supplied list that are directly relevant to answering the question. Select between {min_topics} and {max_topics} topics. Choose exhaustively but do NOT invent new topics. Return the chosen topics exactly as they appear in the list. Always return at least {min_topics} topics. —Instructions— 1. The list of allowed topics will be provided in the placeholder {TOPIC_LIST}. 2. Read the user question provided in the placeholder {question}. 3. Identify every topic from {TOPIC_LIST} that is pertinent to the question. 4. Output only valid JSON. Do not include markdown, comments, or extra text. 5. Output JSON format: { "topics": ["TopicLabel1", "TopicLabel2", ...]} 6. You MUST ONLY choose from the list provided below. Do not invent or rephrase any subtopics. 7. If you cannot find any relevant topics, just find the most relevant {min_topics} topics. —Question— {{question}} —Allowed Topics— {{TOPIC_LIST}}

Table 12: Topic Selection Prompt

A.5 Subtopic Selection Prompt

Instruction description
—Goal— Given the user’s question, choose all subtopics from the supplied list that are directly relevant to answering the question. For the given topic {TOPIC_LABEL}, choose every subtopic from the list below that is helpful for answering the user’s question. Select {min_subtopics} to {max_subtopics} subtopics. Do NOT invent new subtopics. Always return at least {min_subtopics} subtopics, unless the list is shorter than {min_subtopics}. —Instructions— 1. Consider only the subtopics provided in {SUBTOPIC_LIST}. 2. Read the user’s question provided in {question}. 3. Output your selection as valid JSON without markdown, comments, or extra text. 4. Preserve the original order of {SUBTOPIC_LIST} when listing the chosen subtopics. 5. Output JSON Format: {"subtopics": ["SubLb1", "SubLb2", ...]} 6. You MUST ONLY choose from the list provided below. Do not invent or rephrase any subtopics. 7. If you cannot find any relevant subtopics, just find the most relevant {min_subtopics} subtopics. —Question— {{question}} —Allowed Subtopics for {{TOPIC_LABEL}}— {{SUBTOPIC_LIST}}

Table 13: Subtopic Selection Prompt

B Implementation Details

Our implementation details on experiments are as follows:

- NaiveRAG and TH-RAG used Faiss as the vector DB for retrieval.
- For similarity calculation with the query, we did not use Faiss’s built-in L2-distance or inner product but implemented cosine similarity.
- Answer generation prompts were unified across all methods, and the rest of the settings were based on the default values of the respective baselines.
- We fixed the chunk size at 1200 tokens and overlap at 100 tokens for all methods. The temperature during answer generation was set to 0, and gleaning was also set to 0.
- Including graph construction and answer generation, we used GPT-4o-mini from *OpenAI* when needed, and for sentence and chunk embeddings, we used *text-embedding-small-3* from *OpenAI* for all methods.

C Datasets and Baselines Details

C.1 Datasets

- **UltraDomain**: A collection of 20 domain-specific datasets consisting of long-form passages, making it suitable for broad-type evaluation. We generated a total of 125 questions following the methodology of [Edge et al. \(2024\)](#); [Guo et al. \(2024\)](#).
- **HotpotQA**: A Wikipedia-based QA dataset that requires multi-hop reasoning over two to four steps. Each question is paired with context containing both relevant and irrelevant paragraphs. HotpotQA provides two evaluation settings: **Distractor** and **FullWiki**. We adopt the Distractor setting, where 8 out of 10 paragraphs are irrelevant, to better evaluate retrieval accuracy. Each passage is a short Wikipedia paragraph of approximately 100 tokens.
- **MultiHop-RAG**: A QA dataset constructed from English news articles, requiring multi-hop reasoning across 2–4 documents. Question types include **Inference**, **Comparison**, **Temporal**, and **Null**. Each passage contains

2,000–3,000 tokens, making it suitable for long-context evaluation.

C.2 Statistics of KG from TH-RAG

Table 14 presents graph statistics of TH-RAG across the entire dataset.

C.3 Baselines

- **NaiveRAG**: Retrieves the top-8 chunks with the highest similarity to the query. While NaiveRAG typically assumes smaller chunk sizes (e.g., 300 tokens), our setting uses 1,200-token chunks, which may disadvantage its performance. To ensure fairness, we allow NaiveRAG to use 8 chunks, whereas TH-RAG uses 5.
- **GraphRAG**: One of the earliest successful attempts at integrating KG construction into RAG. It provides two configurations: **Global**, which follows the original paper’s method and employs global community summarization, and **Local**, which uses finer-grained communities for answer generation. We evaluate both variants, denoted as **GraphRAG-G** and **GraphRAG-L**. This method uses summarized community reports, triplet information, and chunks to generate answer.
- **LightRAG**: An efficient variant of GraphRAG designed to improve retrieval efficiency. As it is known for its simplicity and scalability, we primarily use it as a baseline for efficiency comparison. This method uses chunks and triplets information for generating answer.
- **PathRAG**: A method specialized for multi-hop reasoning, built on LightRAG. It focuses retrieval on necessary information by connecting entities and pruning paths to the answer. This method uses similar information with LightRAG for answer generation.
- **HyperGraphRAG**: A recent state-of-the-art approach that extends triplet structures to hyperedges, thereby capturing relations among multiple entities simultaneously. This method uses hyperedge-, entity-, and chunk-level information for generating answer.

In addition, other strong baselines exist, such as [Gutiérrez et al. \(2025\)](#); [Zhu et al. \(2025b\)](#); [Zhao et al. \(2025\)](#), as well as chunk-to-graph methods

Category	Agriculture	CS	Legal	Mix	HotpotQA	MultiHop-RAG
Nodes	44,588	45,921	50,162	19,806	50,256	26,250
Topic Nodes	1,568	531	424	401	374	446
Subtopic Nodes	12,280	15,142	14,319	5,993	9,188	7,921
Entity Nodes	30,740	30,248	35,419	13,412	40,694	17,883
Edges	76,946	76,598	94,507	31,580	87,757	42,857
Topic-Subtopic	18,424	20,675	19,212	7,436	12,672	9,825
Subtopic-Entity	35,219	34,017	41,268	14,312	43,843	19,680
Entity-Entity	23,303	21,906	34,027	9,832	31,242	13,352

Table 14: Detailed KG statistics of datasets.

Model	F1	Precision	Recall	Acc
TH-RAG (Main)	0.671	0.692	0.685	0.612
TH-Large	0.631	0.651	0.647	0.587
TH-Large-Tune	0.668	0.689	0.683	0.614

Table 15: Performance on a large-scale integrated corpus (~ 11 M tokens).

like Sarthi et al. (2024); Liu et al. (2025a). We exclude the former because they do not operate on fixed-length chunks, and the latter because they are not based on triplet-style graph construction. Wang et al. (2025) would have been an informative comparison, but the absence of released code made it infeasible to evaluate their method empirically. Moreover, methods not using fixed-length chunks are typically designed for short-passage multi-hop datasets such as HotpotQA, whereas our evaluation covers both long-passage (MultiHop-RAG) and abstractive QA (UltraDomain). For this reason, we select baselines that either share the same purpose as TH-RAG or explicitly aim for global understanding. More experiments can be found on Appendix D.3

C.4 Metrics

We used two evaluation approaches depending on the dataset type.

For the **UltraDomain** dataset, we followed previous studies and used the LLM-as-a-judge method (Zheng et al., 2023). Similar to Guo et al. (2024), answers were compared 1-vs-1 in three dimensions, and the overall win rates were computed. This approach was adopted due to the longer answer nature of this dataset.

For the **MultiHop-RAG** and **HotpotQA** datasets, we adopted traditional evaluation metrics—F1, Recall, Precision, and Accuracy—as the answers are typically short and fact-based. While LLM-as-a-judge has demonstrated strong align-

Method	F1	Precision	Recall	Accuracy
HippoRAG2	0.653	0.685	0.657	0.582
TH-RAG	0.671	0.692	0.685	0.611

Table 16: Comparison of TH-RAG with HippoRAG2 on HotpotQA. TH-RAG clearly outperforms HippoRAG2 across all metrics, highlighting its robustness in both retrieval and answer generation.

ment with human evaluation, it may introduce bias. Therefore, we employed quantitative metrics to provide a more objective assessment of our method on these datasets. For both HotpotQA and MultiHop-RAG, we followed the official evaluation protocol of HotpotQA. Accuracy is determined by whether the predicted answer contains the gold answer.

For retrieval evaluation, we additionally used Recall, F1, Recall@5, and NDCG@5 to ensure a fair and comprehensive comparison. All methods either generate answers from specific chunks or indicate the chunk IDs from which their context is derived; we consider these as the predicted chunks. For the gold chunks, we use those that contain the supporting evidence for each query, treating them as ground truth for retrieval evaluation.

D More Analysis on TH-RAG

D.1 Analysis on Large-scale Corpus

To verify the robustness of TH-RAG in large-scale environments, we conducted additional experiments by constructing a significantly larger and noisier graph. We integrated six distinct datasets (four subsets from UltraDomain, HotpotQA, and MultiHop-RAG) into a single unified corpus, totaling approximately 11M tokens.

We evaluated 1,000 QA pairs from HotpotQA using two variants:

- **TH-Large:** The graph is scaled up, but hy-

hyperparameters remain identical to the main experiments.

- **TH-Large-Tune:** The topic selection ranges are expanded to adapt to the larger scale, specifically increasing $\max(N_T)$ from 10 to 20 and $\max(N_{ST})$ from 25 to 50.

As summarized in Table 15, TH-RAG maintains strong performance even when the graph size increases and includes substantial irrelevant information. While the vanilla TH-Large shows a slight performance dip, adjusting the hyperparameters (*TH-Large-Tune*) effectively recovers the accuracy, demonstrating that the method can flexibly adapt to varying corpus scales.

D.2 Consistency and Convergence of Semantic Topics

While TH-RAG relies on LLM-generated topics and subtopics, our analysis demonstrates that this semantic hierarchy is neither arbitrary nor unbounded, exhibiting both structural consistency and empirical convergence.

Structural Consistency and Alignment. As reported in Table 14, the ratio between topics and subtopics remains remarkably stable across various datasets, consistently falling within the range of 1:10 to 1:30. This suggests that the LLM-generated hierarchy maintains a balanced distribution regardless of the specific corpus.

Furthermore, we assessed the quality of these nodes by comparing them with established ontological structures like Wikidata. For instance, in HotpotQA and MultiHop-RAG, 33 out of 36 generated topics directly matched Wikidata schema categories. Even for unmatched cases (e.g., *concepts*, *lists*), we identified semantically near-equivalent substitutes (e.g., *recognition*, *organizations*), confirming that the generated hierarchy aligns well with human-curated knowledge schemas.

Empirical Convergence in Large-Scale Settings. A critical observation emerged during our scaling experiments (Section D.1) regarding the semantic overlap of topics. If topic generation were entirely stochastic, the number of unique nodes would grow linearly with the corpus size. However, our results show a significant sub-linear growth when merging datasets:

- **Topics (N_T):** The total count decreased from 3,744 (sum of individual datasets) to 2,499 (merged corpus), a 33% reduction.
- **Subtopics (N_{ST}):** The count decreased from 64,843 to 55,832, a 14% reduction.

This inherent convergence indicates that as the corpus size N increases, the effective N_T does not grow unbounded but converges toward a representative semantic set. Since the retrieval complexity of TH-RAG is $O(N_T)$, this property ensures practical efficiency and scalability for massive corpora.

Robustness via Redundant Retrieval. To further mitigate any residual inconsistencies in LLM outputs, TH-RAG’s retrieval design allows for the selection of multiple topics and subtopics (K_1, K_2) for a single query. Rather than enforcing a single, potentially brittle path, this multi-path retrieval ensures flexible coverage of related concepts and stabilizes overall performance against minor variations in the graph schema.

D.3 Comparison on HippoRAG2

As discussed in the Introduction, graph-based RAG aims to achieve two key objectives: (1) answering abstractive QA by providing a global understanding of the corpus, and (2) supporting multi-hop reasoning for specific QA.

While methods such as Wang et al. (2025) and Luo et al. (2025) have demonstrated in their experimental results, HippoRAG (Jimenez Gutierrez et al., 2024; Gutiérrez et al., 2025) shows a different pattern: it achieves very strong performance on HotpotQA but performs poorly on MultiHop-RAG or UltraDomain. This is largely due to its use of extremely small chunk sizes, comparable to single passages in HotpotQA. Such a design is well suited for short-passages multi-hop reasoning datasets (e.g., HotpotQA, MuSiQue, 2WikiMultihopQA), as explained in Appendix C.3, but it fails to generalize to settings with larger chunks (e.g., MultiHop-RAG) or abstractive QA datasets (e.g., UltraDomain). For this reason, we excluded HippoRAG and similar passage-level chunking approaches from our main experiments.

Nevertheless, we acknowledge the importance of comparison with such approaches on their target datasets. Therefore, we additionally report results on HotpotQA against HippoRAG2, under the following setup. We used GPT-4o-mini for NER & Triplet Extraction, text-embedding-3-small for embeddings, and the same answer prompt as TH-RAG, ensuring consistency in experimental conditions. Chunk size was left unchanged. The results are shown in Table 16.

E Examples

E.1 Triplet Extraction Example

Example Input and Output Format
Example 1 —Input— Moscow State University Lomonosov Moscow State University is a coeducational and public research university. ... MSU was renamed after Lomonosov in 1940 and was then known as "Lomonosov University". It also houses the tallest educational building in the world. ... —Output— <pre>{ "triplet": ["Lomonosov Moscow State University", "was renamed after", "Mikhail Lomonosov"], "sentence": "MSU was renamed after Lomonosov in 1940 and was then known as 'Lomonosov University'.", "subject": { "subtopic": "University", "main_topic": "Education" }, "object": { "subtopic": "Person", "main_topic": "Biography" } }</pre>
Example 2 —Input— Under the leadership of student Marcus L. Urann, who created the bylaws and constitution for the organization, the group formed the Lambda Sigma Eta Society . His society was renamed Phi Kappa Phi in 1900, from the letters of the Greek words forming its motto... —Output— <pre>{ "triplet": ["Lambda Sigma Eta Society", "was renamed", "Phi Kappa Phi"], "sentence": "His society was renamed Phi Kappa Phi in 1900, from the letters of the Greek words forming its motto, Philosophía Krateítō Phōtôn - 'Let the love of learning rule humanity.'", "subject": { "subtopic": "Honor society", "main_topic": "Education" }, "object": { "subtopic": "Honor society", "main_topic": "Education" } }</pre>
Example 3 —Input— When music mogul Scooter Braun acquired Swift's longtime label, Big Machine Records , in 2019, he also gained the rights to the master recordings of Swift's first six studio albums. —Output— <pre>{ "triplet": ["Scooter Braun", "acquired", "Big Machine Records"], "sentence": "When music mogul Scooter Braun acquired Swift's longtime label, ...", "subject": { "subtopic": "Person", "main_topic": "Entertainment" }, "object": { "subtopic": "Record label", "main_topic": "Music industry" } }</pre>

Table 17: Example Input and Output Format for Triplet Extraction with Topic. We separate entities, subtopics, and topics for graph construction.

E.2 Qualitative Analysis of Retrieval Performance

To better understand the retrieval mechanism of TH-RAG, we conducted a case study using the HotpotQA dataset. This analysis categorizes instances into successful retrievals (where the system overcomes graph limitations) and failure cases (where structural or granularity issues occur).

Success Cases

Case 1: Implicit Success

Query: “McLemore Avenue is to Booker T. & the M.G.s as what road in the city of Westminster in London is to the Beatles?”

Supporting Fact	Status	Entities Found in Graph
Fact 1: McLemore Avenue	HIT	McLemore Avenue, The Beatles album
Fact 2: Abbey Road	MISS	(No matched node in KG)

Insight: Even when the gold supporting fact is missing in the graph, the **context-expansion** method using semantically similar sentences enables the system to overcome gaps and retrieve the necessary information.

Case 2: Perfect Recall

Query: “Which Istanbul mosque is unique for retaining a Baroque style of architecture, the Bayezid II Mosque or the Nusretiye Mosque?”

Supporting Fact	Status	Matched Subtopics
Fact 1: Bayezid II Mosque	HIT	Architecture / Ottoman Imperial Mosque
Fact 2: Nusretiye Mosque	HIT	Religion / Mosque

Insight: This demonstrates that TH-RAG can infer the necessary topics from the query (e.g., Baroque style) and ensure complete context by subsequently retrieving all appropriate supporting facts.

Failure Cases & Root Cause Analysis

Case 3: Graph Construction Failed

Query: “In what city is the company that Fastjet Tanzania was originally founded as a part of prior to rebranding based?”

Result: FAILED

Root Cause: The entities *Fastjet Tanzania* and *Fly540*, which are central to the supporting facts, are missing from the graph. Despite the relevant sentence being present in another edge, TH-RAG’s retrieval mechanism failed to locate it, even with context expansion.

Case 4: Subtopic Choice Failure

Query: “The 19 high-rise commercial buildings covering 22 acres between 48th and 51st Streets in New York City feature which style of architecture?”

Core Entities	Topic/Subtopic	Status
Art Deco in the United States	Art / Art Deco Style	COVERED
Art Deco examples in NYC	Architecture / Examples, Skyscrapers	COVERED
Rockefeller Center	Architecture / Commercial Complex	NOT SELECTED

Root Cause: The failure stems from *subtopic selection*. The system successfully identified relevant high-level evidence related to architecture and Art Deco, but it failed to activate the specific subtopic *Commercial Complex*, where *Rockefeller Center* was anchored. As a result, the retrieval process did not traverse the critical entity needed to connect the question target (*Rockefeller Center*) with the correct style evidence, which led the model to drift toward an incorrect architectural style.

E.3 Example of Constructed KG and Retrieval Result

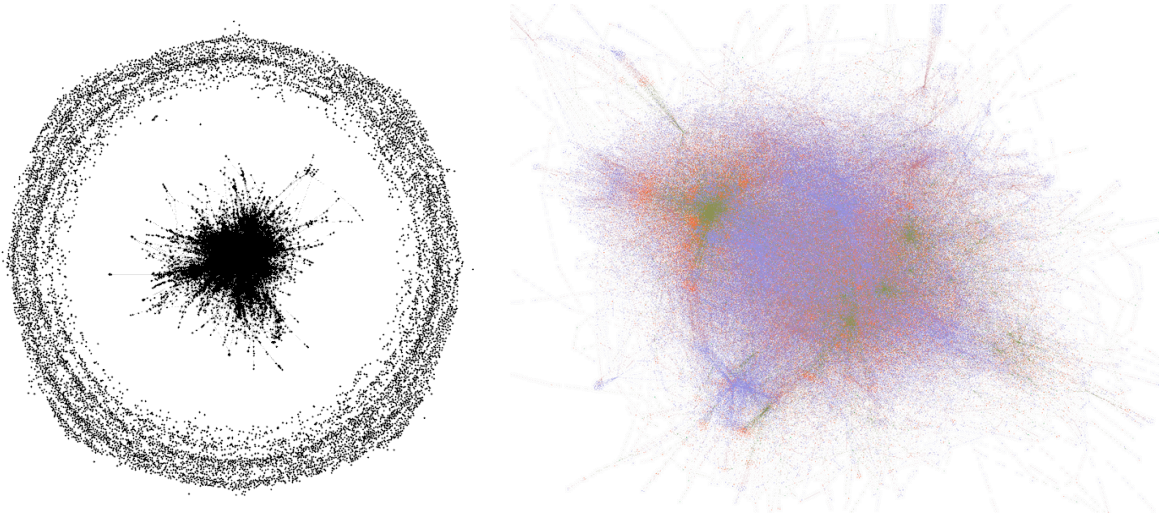


Figure 3: Comparison of KG structures between LightRAG (left) and TH-RAG (right). In the TH-RAG visualization, green nodes represent topics, red nodes represent subtopics, and purple nodes represent entities. The graphs are visualized using the Force Atlas 2 layout algorithm (Jacomy et al., 2014).

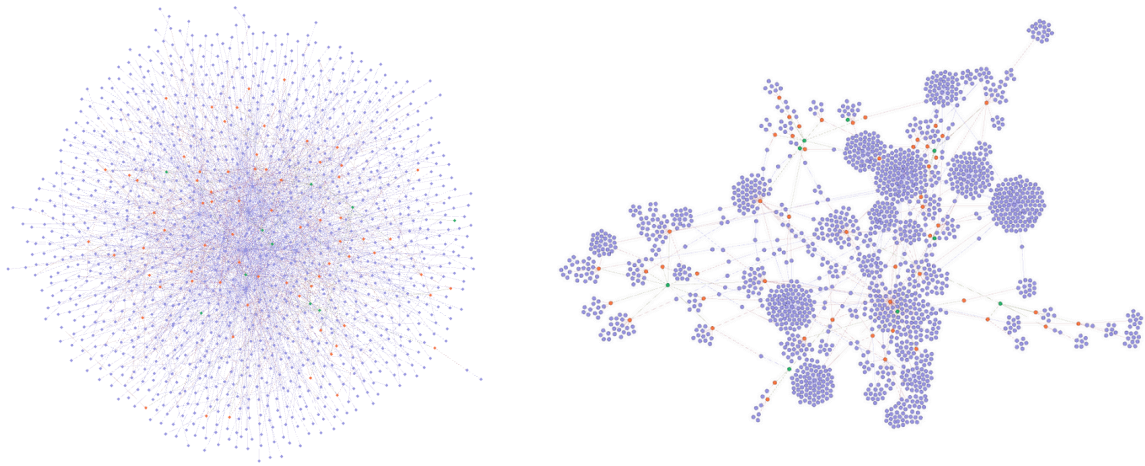


Figure 4: Retrieved subgraphs for different questions using TH-RAG

E.4 Visualization of the Constructed KG with Labels

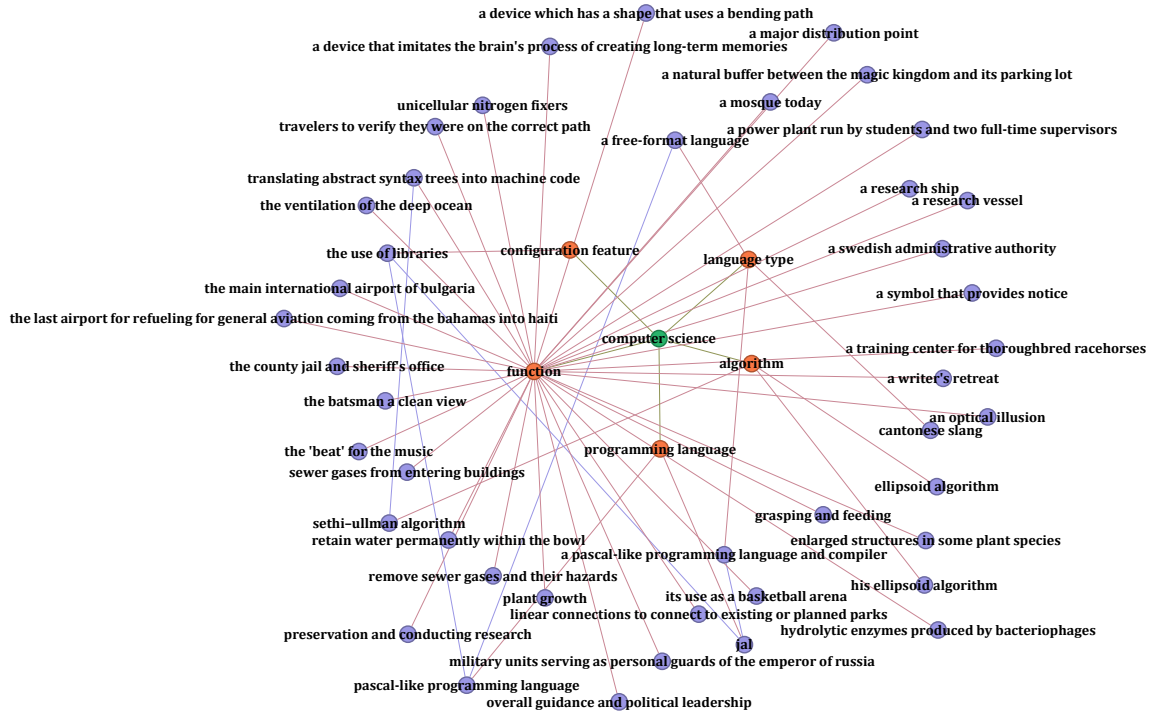


Figure 5: This example shows the topic “computer science” with five related subtopics. For clarity, edges connecting these subtopics to other topics are omitted in the visualization. Although some LLM-generated entities appear overly long or simplistic, TH-RAG is not dependent on their textual quality but instead leverages the connectivity structure of the graph. Moreover, this example illustrates that the topic and its subtopics remain consistent with each other.